United States International University – Africa

DSA 4020 ASSIGNMENT 1

Individual Assignment: Classifying Customer Complaint Tickets

1. Scenario

A Kenyan mobile-money and telecom provider receives thousands of free-text customer complaints every day through SMS, the app and social media. Today, agents read each ticket and route it by hand to one of five departments. Management wants a text classifier that reads a new complaint and routes it automatically.

Your role: you are the junior NLP engineer. You must build, evaluate and critically assess the classifier, then write up your work in a way a non-technical manager could follow.

2. The data

Download the two files provided with this assignment. The data is synthetic (written for teaching), but it imitates real features of Kenyan customer text: spelling mistakes, SMS shorthand, and English mixed with Kiswahili/Sheng.

File	Contents
complaints_train.csv	About 400 labelled tickets: ticket_id, complaint_text, category. Use this for all training, tuning and cross-validation.
complaints_test_unlabelled.csv	About 100 tickets with NO labels: ticket_id, complaint_text. You will predict these and submit your predictions. The lecturer holds the true labels.

The five categories:

Category	Meaning
mpesa_transaction	Failed, reversed, duplicated or mis-sent mobile-money transactions; wrong charges; agent and paybill/till problems.
network_data	Calls, SMS, mobile data bundles, signal, fibre/router problems.
loan_credit	Mobile loans, overdrafts, limits, interest, penalties, credit-bureau listing.
account_access	PIN, OTP, SIM swap, blocked/frozen accounts, login, profile updates.
fraud_scam	Impersonation, fake prize messages, hacking, unauthorised withdrawals, phishing.

Important: some tickets mention more than one issue, and a few labels are debatable (for example a scam that involves a wrong M-Pesa transaction). This is deliberate. Real data is like this, and Part 6 asks you to reason about it.

3. Rules

- This is an individual assignment. Discuss ideas if you wish, but your code, results and write-up must be your own.
- Fix a random seed (random_state=42) so your results can be reproduced.

- **Do not use the test file to make any decision.** Fit every vectoriser, tuning choice and model on the training data only. Predict the test file once, at the end.
- You may use pre-trained embeddings (e.g. GloVe) but no pre-trained classifiers or Transformer models in the main parts. Part 7 (bonus) allows them.
- If you use an AI assistant, state exactly how in a short note at the end. You must be able to explain every line of code you submit.
- Every number or claim in your write-up must be supported by output shown in the document.

4. Tasks

In each task, paste your code (or the key part of it), show the output, and answer the questions in your own words. Use the answer boxes or replace them with your text.

Part A: Understand the data (10 marks)

Q1. Load the training data. Report the number of tickets, the class distribution (table and bar chart) and the average ticket length in words. Is the dataset balanced? **[4 marks]**

Code, output and chart here

Q2. Print three complaint texts from different categories. Identify at least three features that will make classification harder (e.g. typos, shorthand, Kiswahili/Sheng, multiple issues in one ticket). Quote the evidence. **[3 marks]**

Your answer

Q3. Why would a model that always predicts the biggest class look good on accuracy here, and why is that misleading? Compute its accuracy on the training data to prove your point. **[3 marks]**

Your answer

Part B: Preprocessing (20 marks)

Q4. Write a function that lowercases, removes punctuation and numbers, tokenises (NLTK), and optionally removes stop words and applies stemming or lemmatisation. Show a before/after example

for the same ticket under four settings: none, stop words only, stop words + stemming, stop words + lemmatisation. **[6 marks]**

Code and output

Q5. Report vocabulary size (unique tokens) under each of the four settings in a table. Which step shrinks the vocabulary most and why? **[4 marks]**

Table and explanation

Q6. Code-switching. List at least five Kiswahili/Sheng words from the data (for example nimeshindwa, pesa, mkopo). What happens to them when you apply English stop-word removal, stemming and lemmatisation? Give concrete before/after examples and propose two practical ways a Kenyan company could handle this problem. **[6 marks]**

Your answer with examples

Q7. Name one preprocessing step that could HURT this particular task and explain why. (Hint: think about what the tickets contain and which words carry the signal.) Support your argument with a small experiment if you can. **[4 marks]**

Your answer

Part C: Representation (15 marks)

Q8. Build a Bag-of-Words and a TF-IDF representation using TfidfVectorizer / CountVectorizer. Report the matrix shape and sparsity. Show the five highest-weighted terms for each category and comment on whether they make sense. **[5 marks]**

Code, output and comments

Q9. Compare unigrams only with unigrams + bigrams. Give one bigram that is clearly informative for a category and explain why a unigram alone would miss it. **[4 marks]**

Your answer

Q10. Train a Word2Vec model on the training tickets (vector_size 50 or 100) and, if you can, load GloVe vectors. Show the five nearest neighbours of three words of your choice. Convert each ticket to a vector by averaging its word vectors. Explain (a) why a small corpus gives weak embeddings, and (b) what information is lost by averaging. **[6 marks]**

Code, output and explanation

Part D: Modelling (15 marks)

Q11. Using 5-fold stratified cross-validation on the training data only, compare at least three models (e.g. Multinomial Naive Bayes, Logistic Regression, Linear SVM) on TF-IDF features, and at least one model on embedding features. Present mean and standard deviation of macro-F1 in a table. **[6 marks]**

Code and results table

Q12. Run one controlled experiment: change ONLY one thing (for example stemming vs lemmatisation, stop words on/off, or class_weight='balanced'). Report the effect with its standard deviation. Is the difference larger than the noise? Would you claim one option is better? **[5 marks]**

Your answer

Q13. Choose your final model and justify the choice using the evidence above, not just the highest number. Mention speed, simplicity and interpretability. **[4 marks]**

Your answer

Part E: Evaluation and test predictions (20 marks)

Q14. Hold out 20% of the training data (stratified) as a validation set. Train your final model on the remaining 80% and report accuracy, macro precision, macro recall, macro-F1 and weighted-F1. Show the full classification report. **[6 marks]**

Code and output

Q15. Plot the confusion matrix. Which two categories are confused most often? Give one real misclassified ticket and explain, in plain language, why the model made that mistake. **[4 marks]**

Plot and explanation

Q16. fraud_scam is the smallest class, yet missing a fraud ticket may be the most costly error for the company. Which metric for that class (precision or recall) should management watch most closely, and why? Report both for fraud_scam. **[3 marks]**

Your answer

Q17. Retrain your final model on ALL the training data, predict the labels for complaints_test_unlabelled.csv and save them as predictions.csv with exactly two columns: ticket_id, category. Marks are awarded for a valid file (2) and for performance on the hidden labels (5, scaled by macro-F1 against the class average). **[7 marks]**

Show the code you used and the first 10 rows of your predictions

Part F: Reflection and communication (10 marks)

Q18. Write a half-page note to the head of customer care (non-technical) explaining what the classifier can and cannot do, how accurate it is, and what you recommend before it is deployed (e.g. human review of low-confidence tickets, monitoring, retraining). **[4 marks]**

Your note

Q19. List two limitations of this study (data, method or evaluation) and say how you would address each with more time and real data. **[3 marks]**

Your answer

Q20. Ethics and privacy. Real complaint tickets contain phone numbers, names and transaction details. Name two risks of using such data to train a model and two safeguards you would put in place. **[3 marks]**

Your answer

Q21. Fine-tune or apply a pre-trained multilingual Transformer (for example a small DistilBERT-style model) or a sentence-embedding model on the same split. Compare macro-F1, training time and error patterns with your best classical model. Say honestly whether the extra complexity is justified for 400 tickets.

5. Presentation and reproducibility (10 marks)

Marks are given for: a clear structure that follows the numbering above; readable tables and plots with titles and axis labels; code that runs from top to bottom without errors; a fixed random seed; sensible comments; and correct, consistent referencing of any external sources.

6. Marking scheme

Component	Marks	What earns full marks
A. Data understanding	10	Correct statistics, insightful identification of difficulties, sound reasoning on accuracy
B. Preprocessing	20	Working pipeline, clear comparison, thoughtful treatment of Kiswahili/Sheng and harmful steps
C. Representation	15	Correct BoW/TF-IDF/n-grams and embeddings, with interpretation, not only output
D. Modelling	15	Valid cross-validation (training data only), fair comparison, justified choice
E. Evaluation and predictions	20	Correct metrics, confusion analysis, cost-aware reasoning, valid and accurate predictions file

Component	Marks	What earns full marks
F. Reflection and communication	10	Clear, honest, non-technical writing; realistic limitations; ethics
Presentation and reproducibility	10	Professional document, runnable code, fixed seed
Total	100	
Bonus	+5	Well-executed Transformer comparison

Penalties: using the test set to tune or select anything (up to −15); results with no code or output shown (marks lost for that question); late submission as per university policy; copied work (referred under the academic integrity policy).

7. What to submit

1. **Assignment document (.docx):** this document with your code, outputs, plots and answers, named Surname_StudentID_NLP_Assignment.docx
2. **predictions.csv:** two columns, ticket_id and category, one row for each of the tickets in complaints_test_unlabelled.csv
3. **Notebook (.ipynb):** your complete Colab notebook, which must run from top to bottom

AI-use declaration (required):

Tip: do the work in Colab, paste the code and screenshots of the outputs into this document as you go, and finish with the written explanations. Marks are for your reasoning, not just for the code.